

~~Types of machine learning~~

Statistical Learning Theory:

Statistical modeling:

↳ formalization of relationships b/w variables in the form of mathematical equation.

↳ It is the broad framework for studying the concept of inference in both supervised and unsupervised machine learning.

↳ Inference covers the entire spectrum of machine learning from gaining knowledge, making predictions or decisions and constructing model from set of labeled or unlabeled data:

↳ The entire process is stated in statistical framework with every assumption stated mathematically as null or alternative hypothesis.

↳ Statistics is the mathematical study of data.

↳ Using statistics, an interpretable statistical model is created to describe the data, and this model can then be --

↳ The goal of statistical learning theory is to study, in a statistical framework the properties of learning algorithms.

↳ The practical goals of this approach are to make machine learning more precise and to create new or improved modeling algorithms.

↳ This is primarily accomplished by providing a formal, statistical definition of abstract concepts like learning, generalization, overfitting and performance, then these hypotheses one parameter at a time.

↳ The general learning approach is the same as any other scientific discipline.

1. Observe a phenomenon.

2. Construct a model of that phenomenon.

3. Make predictions using this model.

But in statistical machine learning, the entire process needs to be automated for a computer program to learn from it.

So each step of scientific method assumed to be governed by a probabilistic model of the phenomenon. At the simplest level, this means assuming that if all past and future observations are sampled randomly and independently through continuous statistical hypothesis testing, the information about the underlying phenomenon (probability distribution) can be reliably inferred. Just as important, this allows the machine to construct learning algorithms such as k -nearest neighbor with appropriate k .

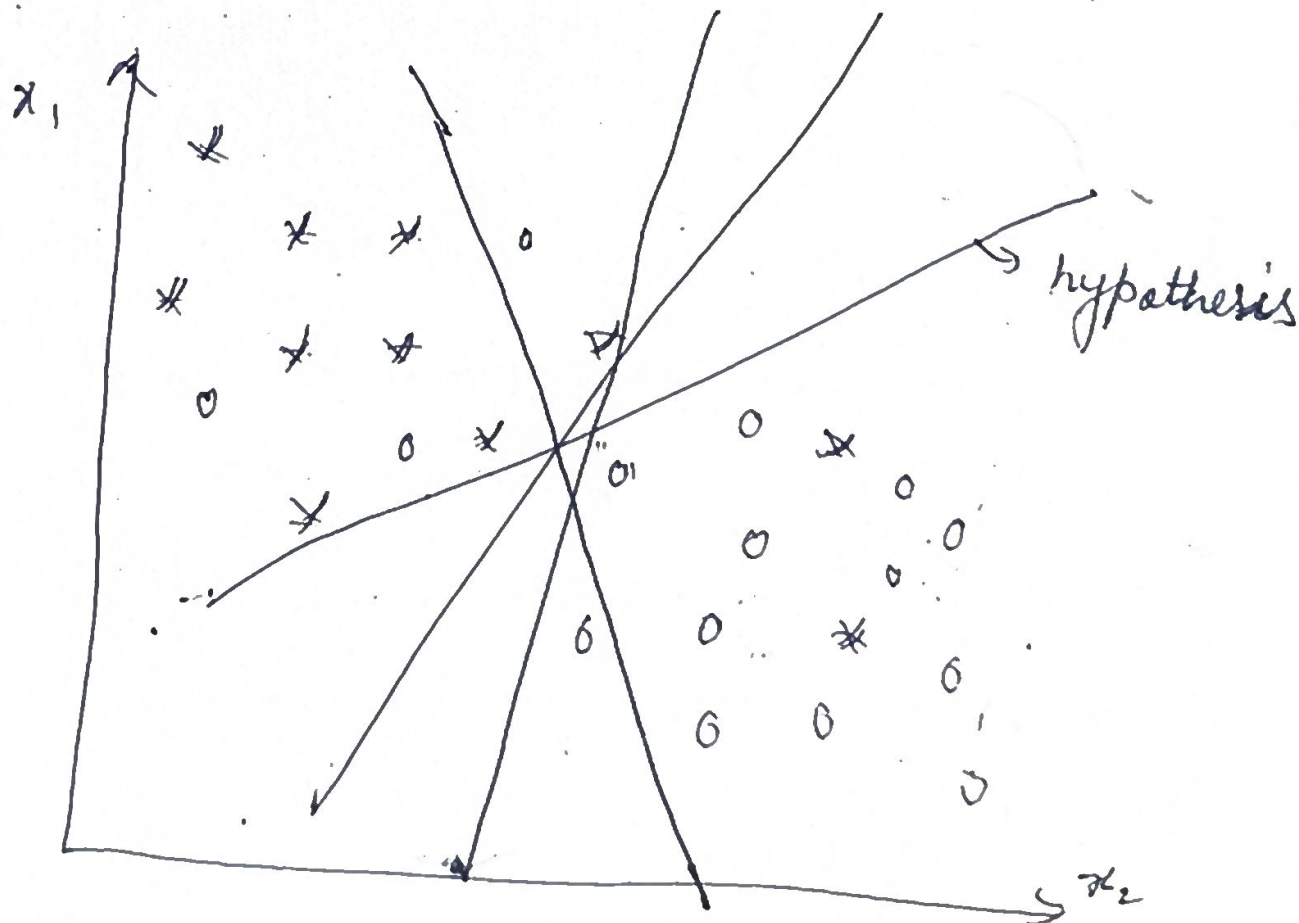
hypotheses space (H)

- ↳ It used by machine learning system is the set of all hypotheses that might possibly be return by it
- ↳ It is the set of all possible legal hypotheses.
- ↳ This is the set from which the machine learning algorithm would determine the best possible (only one) which would best describe the target function or the outputs.

Hypothesis (h)

↳ A hypothesis is a function that best describes the target in supervised machine learning.

- ↳ The hypothesis that an algorithm would come up depends upon the data and also depends upon the restriction and bias that we have imposed on the data.



Statistical learning theory (SLT)

- ↳ It is mathematical foundation of machine learning.
- ↳ It studies how and why learning algo can generalize from a limited set of training examples to unseen data.
- ↳ It is developed by ~~val~~ Vladimir Vapnik
- ↳ It provides theoretical guarantees about a model performance rather than focusing only on empirical accuracy.

In supervised learning

$$D = \{(x_1, y_1), (x_2, y_2) \dots (x_m, y_m)\}$$

where x_i = input features

y_i = target labels.

The goal is to learn a function $f(x)$ that predicts the correct output for new, unseen examples.

Statistical learning theory ask

- How much training data is enough
- why do some model generalize better than others
- How can we avoid overfitting
- what is trade-off b/w model complexity & prediction accuracy

1. Risk (Expected Error)

True error of a model is called risk.

$$R(f) = E[L(y, f(x))]$$

where L = loss function.

- E = expectation over the unknown data distribution.

2. Empirical Risk.

We estimate the error using the training data

$$R_{\text{emp}}(f) = \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i))$$

This is called empirical risk or training error

3. Empirical Risk Minimization (ERM)

$$f^* = \underset{f}{\operatorname{argmin}} R_{\text{emp}}(f)$$

However, minimizing training error alone can lead to overfitting.

4. Overfitting vs Underfitting

• low training error

• high test error

• Model memorize training data

→ high training error

→ high test error

→ Model is too simple.

↳ It provides a robust set of tools for understanding patterns and trend and making inferences and predictions based on data.

↳ when dealing with large dataset, statistics helps us understand and summarize the data, allowing us to make sense of complex phenomena.

↳ The principles of statistics are the very pillars that uphold the structure of machine learning.

↳ Constructing machine learning models. ex linear regression

↳ Interpreting results. ex P-value, R-squared value

↳ Validating models. ex hypothesis testing, cross validation

↳ Underpinning advance technique

ex gradient descent in neural networks

ML: An algorithm that can learn from data without relying on rule-based programming.

Statistical Modeling: formalization of relationship between variables in the form of mathematical equation

• It usually work with a number of assumptions.

SLT is a theoretical framework for machine learning that use statistics and function analysis to understand how algorithms can learn and generalize from empirical data.

→ Statistical learning theory is framework for machine learning drawing from the fields of statistics and functional analysis.

→ It deals with statistical inference problems of finding a predictive function based on data.

→ From perspective of statistical learning theory, supervised learning is best understood.

→ Depending on the type of output supervised learning problems are

↳ regression ex $V = \mathbb{R}$, $0/p$ continuous

↳ classification ex face recognition
 $0/p$ categorical.

↳ By employing statistical methods, we can extract significant patterns, relationships and insights from intricate datasets.

↳ Thus, promoting the effectiveness machine learning algorithms.

Role of ^{statistics} statistics in Machine learning

↳ It is backbone of ML, providing the tools and techniques to analyse and interpret data.

↳ Statistics is the science that allows us to collect, analyse, interpret, present and organize data.

VC Dimension

↳ one of most important idea in SLT.

↳ It measures the capacity of learning model.

Ex Constant classifier = VC Dimension = 1
Linear " in 2D " " = 2
Deep NN " = very high VC Dim.

Higher VC dimension means.

- More flexibility
- Greater ability to fit data
- Higher risk of overfitting

~~Bias-Variance~~
Learning process ↓
Training Data

↓
Choose hypothesis space

↓
Train model

↓
Evaluate Generalization

↓
Prediction on new data